

ANDERSON JUNIOR COLLEGE
JC2 Preliminary Examinations 2013

MATHEMATICS

8864/01

Higher 1
Paper 1

17 September 2013
3 hours

Additional Materials: Graph Paper
List of Formulae

READ THESE INSTRUCTIONS FIRST

Write your name, PDG & index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or liquid fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the lesson, fasten your work securely together.

The number of marks is given in [] at the end of each question or part question

Name _____

PDG _____

1	2	3	4	5	6	7
8	9	10	11	12	Total	

This Question Paper consists of 6 printed pages.

[Turn over]

Section A: Pure Mathematics [35 marks]

1. Find

(i) $\int_1^3 \frac{4}{(3x-1)} dx$, expressing your answer in the form $A \ln B$, where B is a prime number. [2]

(ii) $\int (1 - e^{-2x})^2 dx$ [2]

2. (i) Find algebraically in exact form, the set of values of p for which $(p-1)x^2 - 3x + p + 1 > 0$ for all real values of x . [4]

(ii) Hence find the values of p for which $y = 2(p-1)x^3 - 9x^2 + 6(p+1)x + p$ is a strictly increasing function of x . [2]

3. Sketch the graph of $y = 4k^2 - x^2$, where k is a positive constant, showing clearly the coordinates of the points where the curve cuts the axes. [2]

Insert on your sketch, the graph of $y = 3x^2 + 3k^2$. [1]

Find in terms of k , the x -coordinates of the points where the two graphs intersect. [1]

Hence find, in terms of k , the area of the finite region enclosed by the two curves. [3]

4. (i) Sketch on the same diagram, the graphs of $y = 2e^{2x} - 1$ and $y = 3e^{-2x}$, showing clearly the equations of any asymptotes and the coordinates of the points where the curve crosses the y -axis. [2]

Find the exact x -coordinate(s) of the point(s) where the graphs of $y = 2e^{2x} - 1$ and $y = 3e^{-2x}$ intersect. [4]

(ii) Use your graph to deduce in exact terms, the values of x that satisfy the inequality $2e^{2x} - 3e^{-2x} - 1 > 0$, showing your reasons clearly. [2]

(iii) Deduce the values of t for which $2e^{-t} - 3e^t - 1 > 0$. [2]

5. The equation of a curve C is given by $y = x^2 - \ln(2x - 1)$ for $x > \frac{1}{2}$.
- (i) Find by differentiation, the x -coordinate of the turning point of C .
Determine the nature of this turning point. [4]
- (ii) P is the point on the curve C for which $x = 2$. The normal to C at P meets the x -axis at A .
Find the exact x -coordinate of A , simplifying your answer. [4]

Section B: Statistics [60 marks]

6. In Super Junior College (SJC), there are 850 male students, 900 female students and 250 staff. Best Surveyor Company (BSC) plans to collect data on 40 members of SJC to estimate the number of hours spent on social networking sites by a typical member of SJC. Knowing that reporting time of SJC is 730 am and that the college gate opens at 630am, BSC deploys employees to station at the gate of SJC at 7am to survey the first 17 male students, first 18 female students and first 5 staff that they see.
- (i) State, in this context, one disadvantage of BSC's sampling method. [1]
- (ii) State a more appropriate sampling method that BSC might use in this case, explaining clearly how this is an improvement from BSC's method. [2]
7. The heights of a certain species of plant at maturity is known to have a mean of 100 cm and standard deviation of 5 cm.
- (i) Gardener Jeremy planted 60 seedlings and measured them upon maturity. Find the probability that the average height of his plants is less than 99.1 cm. [2]
- (ii) A large number of gardeners take part in a competition. Each gardener planted 60 seedlings and measured them upon maturity. Gardeners whose plants have an average height of more than k cm will be awarded a prize. If 5% of the gardeners are awarded a prize, find the minimum value of k . [2]

8. In a volleyball match which comprises of 3 games, the probability that Team Feoniz wins the first set is 0.6. If a game is won, the probability of Team Feoniz winning the subsequent game is increased by 0.1. If a game is lost, the probability of Team Feoniz losing the subsequent game increases by 0.1. The team that first won two games wins the match. Each game results in either a win or a loss for Team Feoniz.
- (i) Show that the probability that Team Feoniz wins the match is 0.648. [2]
 - (ii) Given that Team Feoniz wins the match, find the probability that it does so in the third game. [3]
 - (iii) Team Feoniz played a total of three matches in the Preliminary stage. The result of a match is independent of the results of any other matches. Given that it wins all three matches, find the probability that only one of the matches was won in the third game. [3]

9. A teacher of Peaceful Childcare Centre claims that a randomly chosen child spends 4 hours on afternoon nap. A survey was conducted and the time, x hours, spent by 50 randomly chosen children on afternoon naps is as follows:

$$\sum x = 185, \quad \sum (x - \bar{x})^2 = 39.2$$

- (i) Find an unbiased estimate of the population mean and variance. [2]
 - (ii) Carry out a test at 2% level of significance to determine if the teacher's claim is correct. [4]
 - (iii) Suppose the teacher now claims that a randomly chosen child spends less than 4 hours on afternoon nap. Without carrying out another test, state, with a reason, whether the conclusion in part (ii) would remain the same. [2]
 - (iv) The principal of Peaceful Childcare Centre now claims that a randomly chosen child spends μ_0 hours on afternoon nap. Find the set of values of μ_0 such that the principal is confident, at 5% level of significance, that he did not understate the mean time that the children took for afternoon naps. [4]
10. Epple Company manufactures a large number of E-phones. It is known that 1% of the E-phones manufactured are defective.
- (i) A random sample of n phones was selected. Using an algebraic method, find the smallest value of n such that the probability that there is at least one defective phone in the sample is more than 0.95. [3]
 - (ii) A carton, which consists of 24 E-phones, will be rejected if there are at least two defective phones. Show that the probability that a randomly chosen carton is being rejected is 0.0239. [2]
 - (iii) Moon-hub retailer ordered a total of 250 cartons of E-phones. By using a suitable approximation, find the probability that between five and ten cartons of E-phones will be rejected. [4]

11. In Sunny Hotel, the weight, in kg, of a randomly selected tourist follows independent normal distributions with means and standard deviations as shown in the table below:

	Mean	Standard deviation
Male tourist	75	10
Female tourist	μ	σ

Given that 10% of the females weigh less than 43kg and 18% of the females weigh more than 55kg, show that $\mu = 50$ and find the value of σ . [3]

- (a) Five female tourists and one male tourist are selected at random. Find the probability that
- (i) their mean weight exceeds 60kg [3]
 - (ii) the total weight of the female tourists differs from thrice the weight of the male tourist by more than 5kg. [3]
 - (iii) half of them weigh more than 55kg [3]
- (b) There are twice as many male tourists as female tourists.
A randomly selected tourist weighs more than 55kg. Find the probability that this tourist is a female. [3]

12. Due to a recent flu epidemic, there has been a spike in the number of face masks and bottles of Vitamin C sold in Country X. A medical supplies company wishes to investigate the relationship between the weekly sales of face masks, f thousands, and bottles of Vitamin C, v hundred bottles in the country. Sales figures were collected over 6 weeks.

f	0.5	4.0	5.9	3.1	1.9	1.1
v	1.0	5.0	7.2	3.6	2.6	1.6

- (i) Give a sketch of a scatter diagram for the data. [1]
- (ii) Find the product moment correlation coefficient and comment on its value in the context of this question. [2]
- (iii) Find the equation of the regression line of v on f , in the form $v = kf + c$, giving your answers to 3 decimal places.
Explain what the value of k means in the context of this question.
Sketch this line on your scatter diagram. [3]
- (iv) Give an estimate of the number of Vitamin C bottles sold on the week when 2500 face masks are sold. Explain whether you would expect this estimate to be reliable. [2]
- (v) The medical officer used the regression line of v on f to estimate the number of face masks sold on a week when 500 bottles of Vitamin C tablets are sold. Comment on the validity of this estimate. [1]